

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, C++, PySpark, SQL

ML Frameworks: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers

MLOps/Cloud: AWS SageMaker, Docker, Kubernetes, FastAPI, Git, CI/CD, MLflow

Data Engineering: Hadoop, Hive, Spark, NiFi, Airflow, ETL Pipelines

Concepts: Deep Learning, Perception, Segmentation, Prediction, Time Series, Computer Vision, LLMs, Reinforcement Learning, Active Learning, MLOps, Edge AI

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Architected and evaluated deep learning models for metrological applications, enhancing model performance and training efficiency.
- Explored novel architectures and physics-informed hybrid loss functions to improve model accuracy and stability in complex systems.
- Collaborated with PhD researchers on a publication for the FLOMEKO conference, contributing to scientific knowledge transfer.
- Developed interactive dashboards for data visualization and real-time model performance monitoring, utilizing data science methods.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Enhanced vitiligo segmentation accuracy to 94% and increased inference speed by 5x using advanced AI algorithms; deployed with Docker and FastAPI.
- Pioneered a real-time tennis foul detection system leveraging TensorRT, reducing compute cost and inference time by 6x for edge deployment.
- Managed and optimized MLOps pipelines on AWS SageMaker, proactively mitigating YOLO model data drift and maintaining Kubernetes infrastructure to ensure robust deployment.
- Developed Proof-of-Concepts (POCs) for client demos using CNNs, YOLO, and few-shot learning techniques, showcasing innovative AI solutions.
- Conducted comprehensive educational sessions on machine learning and big data concepts at universities, fostering knowledge transfer.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed an AI chatbot application using the OpenAI API and relevant LLM tools/frameworks, integrating modern ML techniques.

- Contributed to full-stack development of an AI-based services website using Flask, HTML, and CSS.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Designed and implemented automated web scrapers and ETL pipelines for structured and unstructured data, optimizing data ingestion workflows.
- Automated data cleaning and transformation workflows, significantly improving data quality and efficiency for client analytics use cases.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Architected a weighted-probability ensemble integrating DPFunc and DeepGOPlus, leveraging AlphaFold for robust feature extraction and taxonomic bias mitigation.
- Increased model accuracy by 4% through custom feature engineering across MF, CC, and BP namespaces, demonstrating strong analytical skills.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Developed a real-time Bitcoin price forecasting system utilizing advanced time series prediction models and ML techniques.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Collaborated to build the world's first AI-driven self-playing Chitralli Sitar, combining signal processing and music generation LLMs.
- Project featured on Coke Studio in a music video, showcasing innovative application of AI and public impact.

Hugging Face

Open-Source

Contributor

- Co-authored the Zero-shot Image Classification technical documentation and implementation page on the Hugging Face Hub, facilitating open-source knowledge sharing.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Specializing in Deep Learning, Computer Vision, and Robotics.

National University of Computer and Emerging Sciences

Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor (Fall 2019, Spring 2020, Fall 2020)
- Relevant Coursework: Machine Learning, Deep Learning, Data Structures, Algorithms, Calculus, Linear Algebra

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python